Blood Donation Network API

1. Overview

The API is the **source of truth** for the entire application. The frontend only displays what the API returns and never duplicates business rules.
The backend decides:

- which donor blood types are compatible with a blood request (R-1, R-2)
- whether a donor is in their 56-day post-donation rest period and their exact nextAllowedDonationDate (R-3, R-4, R-5)
- whether a donor is eligible for a request (ELIGIBLE, REST_PERIOD, UNAVAILABLE, INACTIVE) and the exact rejection reason when they are not (R-6)
- the ranked order of eligible donors and the excluded-donor summary counts (restPeriod, unavailable, inactive) for any open request (R-7, R-8)
- the request lifecycle status (OPEN, FULFILLED, CANCELLED) and unitsStillNeeded (unitsRequired - unitsCollected) (R-9, R-10, R-11, R-14)
- whether a pledge is accepted or rejected (at most one active PLEDGED pledge per donor, and active pledges on a request never exceed unitsStillNeeded) (R-15, R-16, R-17)
- atomic recording of a completed donation inside a single database transaction (prisma.$transaction), automatically updating unitsCollected, completing the donor's pledge, starting the donor's 56-day rest period, and—when unitsCollected === unitsRequired—marking the request FULFILLED and releasing all remaining PLEDGED pledges to RELEASED (R-18, R-19, R-20, R-21, R-22)
- privacy filtering on public endpoints so donor personal data, patient references, and coordinator notes are never exposed to the public (R-23, R-24)

The API uses **Node.js**, **NestJS**, **TypeScript**, **PostgreSQL** through **Prisma ORM**, **JWT authentication**, and **Swagger / OpenAPI**. All data persists across server restarts.

2. Base URL & Authentication

```
http://localhost:3000/api
```

Swagger / OpenAPI interactive documentation:
`http://localhost:3000/api/docs`

Frontend environment variable:
`VITE_API_URL=http://localhost:3000/api`

All routes below start with /api.

Authentication & Role Guards

Protected endpoints require a valid JWT in the Authorization header:
`Authorization: Bearer <accessToken>`

Guard / Decorator	Behavior	Error Status & Message
@Public()	Open to everyone without a token. Returns public fields only.	—
JwtAuthGuard	Verifies the Bearer token and checks that the user exists and isActive === true.	401 Unauthorized — { "message": "Authentication required. Please sign in." }
RolesGuard (@Roles('COORDINATOR') or @Roles('DONOR'))	Verifies that req.user.role matches the required role.	403 Forbidden — { "message": "You do not have permission to perform this action." }

3. Endpoint Summary

Method	Endpoint	Access / Role	Purpose	Backend Owner
GET	/areas	Public	List all supported areas	Eyad
POST	/auth/sign-up	Public	Register a new volunteer donor account	Eyad
POST	/auth/sign-in	Public	Sign in as Coordinator or Donor and receive a JWT	Eyad
GET	/auth/me	Authenticated (DONOR or COORDINATOR)	Get current signed-in user profile	Eyad
GET	/requests/sos	Public	List OPEN requests for the public SOS board (public fields only)	Adham
POST	/requests	COORDINATOR	Publish a new OPEN SOS blood request	Adham
GET	/requests	COORDINATOR	List all blood requests with optional filters (status, bloodType, areaId, urgency)	Adham
GET	/requests/:id	COORDINATOR	Get full details of one request (counters, pledges, donations)	Adham
PATCH	/requests/:id	COORDINATOR	Update an OPEN request's unitsRequired,	Adham

Method	Endpoint	Access / Role	Purpose	Backend Owner
			urgency, and note	
PATCH	/requests/:id/cancel	COORDINATOR	Cancel an OPEN request with a reason and release active pledges	Adham
GET	/requests/:id/eligible-donors	COORDINATOR	Get ranked eligible donors and excluded-donor summary by reason	Adham (calls Eyad's MatchingService)
POST	/requests/:id/donations	COORDINATOR	Record a completed 1-unit donation in a database transaction	Adham (calls Eyad's MatchingService)
GET	/donors/me	DONOR	Get own donor profile, eligibility status, next allowed date, active pledge, and donation history	Laith (calls Eyad's MatchingService)
PATCH	/donors/me	DONOR	Update own areaId, phone, and isAvailable flag	Laith
PATCH	/donors/me/deactivate	DONOR	Deactivate own account (isActive = false) and withdraw active pledge	Laith
GET	/donors/me/eligible-requests	DONOR	List OPEN requests the signed-in donor is currently eligible to help with	Laith (calls Eyad's MatchingService)
GET	/donors	COORDINATOR	List all donors with their eligibility status and matching data	Laith
POST	/requests/:id/pledges	DONOR	Pledge to donate to an eligible OPEN request	Laith (calls Eyad's MatchingService)
PATCH	/pledges/:id/withdraw	DONOR	Withdraw own active PLEDGED pledge	Laith

4. Data Shapes

All JSON fields use camelCase.

```
date          YYYY-MM-DD                  example: 2026-12-12
timestamp     ISO 8601 UTC                example:
2026-11-01T09:10:00.000Z
bloodType     'O-' | 'O+' | 'A-' | 'A+' | 'B-' | 'B+' | 'AB-' | 'AB+'
urgency       'CRITICAL' | 'HIGH' | 'NORMAL'
```

4.1 Area

```
{
  "id": "area-homs",
  "name": "Homs"
}
```

4.2 AuthUser & AuthResponse

```
{
  "accessToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "user": {
    "id": "u-donor-sara",
    "phone": "0933112233",
    "email": "sara@example.com",
    "role": "DONOR",
    "isActive": true,
    "donorProfile": {
      "id": "dp-sara",
      "fullName": "Sara Al-Khatib",
      "bloodType": "A-",
      "area": { "id": "area-homs", "name": "Homs" },
      "isAvailable": true
    }
  }
}
```

For a COORDINATOR, donorProfile is null.

4.3 PublicSosRequest (Public SOS Board — FR-8, R-23)

Contains **only** public fields. Never includes patientReference, note, cancelReason, createdBy, pledges, or donations.

```
{
  "id": "req-101",
  "bloodType": "A-",
```

```
  "area": { "id": "area-homs", "name": "Homs" },
  "urgency": "CRITICAL",
  "hospital": "Central Blood Bank Homs",
  "unitsStillNeeded": 3,
  "createdAt": "2026-11-01T09:10:00.000Z"
}
```

4.4 CoordinatorRequestSummary (GET /requests item)

```
{
  "id": "req-101",
  "patientReference": "HOMS-ICU-402",
  "bloodType": "A-",
  "unitsRequired": 3,
  "unitsCollected": 1,
  "unitsStillNeeded": 2,
  "area": { "id": "area-homs", "name": "Homs" },
  "urgency": "CRITICAL",
  "status": "OPEN",
  "hospital": "Central Blood Bank Homs",
  "note": "Emergency surgery in Ward 3.",
  "cancelReason": null,
  "activePledgesCount": 1,
  "createdAt": "2026-11-01T09:10:00.000Z",
  "updatedAt": "2026-11-01T11:30:00.000Z"
}
```

4.5 CoordinatorRequestDetails (GET /requests/:id)

Extends CoordinatorRequestSummary with pledges and donations (FR-13):

```
{
  "id": "req-101",
  "patientReference": "HOMS-ICU-402",
  "bloodType": "A-",
  "unitsRequired": 3,
  "unitsCollected": 1,
  "unitsStillNeeded": 2,
  "area": { "id": "area-homs", "name": "Homs" },
  "urgency": "CRITICAL",
  "status": "OPEN",
  "hospital": "Central Blood Bank Homs",
  "note": "Emergency surgery in Ward 3.",
  "cancelReason": null,
  "activePledgesCount": 0,
  "createdAt": "2026-11-01T09:10:00.000Z",
  "updatedAt": "2026-11-01T11:30:00.000Z",
```

```
  "pledges": [
    {
      "id": "plg-501",
      "status": "COMPLETED",
      "createdAt": "2026-11-01T09:45:00.000Z",
      "donor": {
        "id": "dp-sara",
        "fullName": "Sara Al-Khatib",
        "phone": "0933112233",
        "bloodType": "A-",
        "area": { "id": "area-homs", "name": "Homs" }
      }
    }
  ],
  "donations": [
    {
      "id": "don-901",
      "donationDate": "2026-11-01",
      "units": 1,
      "createdAt": "2026-11-01T11:30:00.000Z",
      "donor": {
        "id": "dp-sara",
        "fullName": "Sara Al-Khatib",
        "phone": "0933112233",
        "bloodType": "A-"
      },
      "recordedBy": {
        "id": "u-coord-1",
        "phone": "0911000001",
        "email": "coordinator.homs@bloodnet.sy"
      }
    }
  ]
}
```

4.6 EligibleDonorsResponse (GET /requests/:id/eligible-donors — FR-9, R-7, R-8)

```
{
  "requestId": "req-101",
  "bloodTypeNeeded": "A-",
  "area": { "id": "area-homs", "name": "Homs" },
  "eligibleDonors": [
    {
      "rank": 1,
      "id": "dp-sara",
```

```
      "fullName": "Sara Al-Khatib",
      "phone": "0933112233",
      "bloodType": "A-",
      "isExactBloodMatch": true,
      "area": { "id": "area-homs", "name": "Homs" },
      "lastDonationDate": null,
      "hasActivePledgeForThisRequest": false
    },
    {
      "rank": 2,
      "id": "dp-omar",
      "fullName": "Omar Mansour",
      "phone": "0944556677",
      "bloodType": "O-",
      "isExactBloodMatch": false,
      "area": { "id": "area-homs", "name": "Homs" },
      "lastDonationDate": "2026-06-10",
      "hasActivePledgeForThisRequest": false
    }
  ],
  "excludedSummary": {
    "restPeriod": 1,
    "unavailable": 1,
    "inactive": 0
  }
}
```


4.7 DonorDashboardProfile (GET /donors/me — FR-3, FR-4, R-24)


```
{
  "id": "dp-sara",
  "userId": "u-donor-sara",
  "fullName": "Sara Al-Khatib",
  "phone": "0933112233",
  "email": "sara@example.com",
  "bloodType": "A-",
  "area": { "id": "area-homs", "name": "Homs" },
  "isAvailable": true,
  "isActive": true,
  "lastDonationDate": "2026-11-01",
  "nextAllowedDonationDate": "2026-12-27",
  "eligibilityStatus": "REST_PERIOD",
  "activePledge": null,
  "donationHistory": [
    {
      "id": "don-901",
      "donationDate": "2026-11-01",
```

```
      "units": 1,
      "request": {
        "id": "req-101",
        "bloodType": "A-",
        "hospital": "Central Blood Bank Homs",
        "area": { "id": "area-homs", "name": "Homs" }
      }
    }
  ]
}
```

When the donor has an active PLEDGED pledge, activePledge has the shape:

```
{
  "id": "plg-501",
  "status": "PLEDGED",
  "createdAt": "2026-11-01T09:45:00.000Z",
  "request": {
    "id": "req-101",
    "bloodType": "A-",
    "area": { "id": "area-homs", "name": "Homs" },
    "urgency": "CRITICAL",
    "hospital": "Central Blood Bank Homs",
    "unitsRequired": 3,
    "unitsCollected": 0,
    "unitsStillNeeded": 3,
    "status": "OPEN"
  }
}
```

4.8 EligibleOpenRequest (GET /donors/me/eligible-requests item — FR-10)

```
{
  "id": "req-101",
  "bloodType": "A-",
  "area": { "id": "area-homs", "name": "Homs" },
  "urgency": "CRITICAL",
  "hospital": "Central Blood Bank Homs",
  "unitsRequired": 3,
  "unitsCollected": 0,
  "unitsStillNeeded": 3,
  "activePledgesCount": 1,
  "canPledge": true,
  "createdAt": "2026-11-01T09:10:00.000Z"
}
```

5. GET /areas

Returns the fixed list of supported areas ordered alphabetically by name.

```
GET /api/areas

200 OK    → Area[]
```

Public endpoint (@Public()). Used by the Donor Sign-Up page, Donor Profile page, and Coordinator Request filters/creation form.

6. Authentication Endpoints (/auth)

Owner: **Eyad Ibrahim**.

6.1 POST /auth/sign-up

Registers a new volunteer donor account, hashes the password with bcrypt, creates both the User (role: 'DONOR') and DonorProfile in one Prisma transaction, and returns an AuthResponse (201 Created).

```
POST /api/auth/sign-up
Content-Type: application/json

{
  "fullName": "Sara Al-Khatib",
  "phone": "0933112233",
  "email": "sara@example.com",
  "password": "DonorPassword123",
  "bloodType": "A-",
  "areaId": "area-homs"
}
```

(email is optional; if provided, it is trimmed and lowercased.)

Checks (in this exact order)

Part 1 — Format (SignUpDto):

1. fullName is a non-empty string after trimming
2. phone is normalized (normalizePhone) and matches @IsValidPhone()
3. email (if provided) is a valid email address
4. password is a string of at least 8 characters
5. bloodType is a valid blood type (@IsValidBloodType(): 'O-' | 'O+' | 'A-' | 'A+' | 'B-' | 'B+' | 'AB-' | 'AB+')
6. areaId is a non-empty string

Part 2 — Rules (AuthService.signUp):

1. areaId exists in the Area table (400)

2. phone is not already registered (409)
3. email (if provided) is not already registered (409)

Errors

Problem	Status	Message
Missing fullName	400	Full name is required.
Missing or invalid phone	400	A valid phone number is required.
Invalid email format	400	Please provide a valid email address.
password shorter than 8 characters	400	Password must be at least 8 characters long.
Missing or unknown bloodType	400	Invalid blood type. Must be one of O-, O+, A-, A+, B-, B+, AB-, AB+.
Missing areaId	400	Area is required.
areaId does not exist in DB	400	Selected area is not supported.
phone already registered	409	An account with this phone number is already registered.
email already registered	409	An account with this email address is already registered.

6.2 POST /auth/sign-in

Authenticates a COORDINATOR or DONOR using identifier (phone number or email) and password.

```
POST /api/auth/sign-in
Content-Type: application/json

{
  "identifier": "0933112233",
  "password": "DonorPassword123"
}

200 OK    → AuthResponse (section 4.2)
```

Per Section 13 ("Sign-in with a wrong password — Rejected as unauthorized, without saying which part was wrong"), a wrong identifier and a wrong password return the **exact same 401 message**.

Problem	Status	Message
Missing identifier	400	Phone number or email is required.
Missing password	400	Password is required.
Unknown identifier or wrong password	401	Invalid credentials. Please check your phone/email and password.
Account deactivated (isActive	403	This account has been

Problem	Status	Message
=== false)		deactivated.

6.3 GET /auth/me

Returns the AuthUser object for the currently authenticated JWT token.

```
GET /api/auth/me
Authorization: Bearer <accessToken>

200 OK    → AuthUser
```

7. GET /requests/sos (Public SOS Board)

Owner: **Adham Albasha**.
Returns all OPEN blood requests ordered by:

1. urgency: CRITICAL first, then HIGH, then NORMAL
2. createdAt: oldest first (ASC)

```
GET /api/requests/sos

200 OK    → PublicSosRequest[] (section 4.3)
```

- Public endpoint (@Public()).
- Enforces **FR-8** and **R-23**: returns **only** id, bloodType, area, urgency, hospital, unitsStillNeeded, and createdAt.
- When there are no OPEN requests, returns 200 OK with [] (Section 13: "The SOS board when there are no open requests — Empty state, not an error").

8. POST /requests (Publish an SOS Request)

Owner: **Adham Albasha**. Role: COORDINATOR.
Creates a new blood request with status: 'OPEN' and unitsCollected: 0 (Flow A, FR-5).

```
POST /api/requests
Authorization: Bearer <coordinatorToken>
Content-Type: application/json

{
  "patientReference": "HOMS-ICU-402",
  "bloodType": "A-",
  "unitsRequired": 3,
  "areaId": "area-homs",
  "urgency": "CRITICAL",
  "hospital": "Central Blood Bank Homs",
  "note": "Emergency surgery in Ward 3."
}
```

Success

```
201 Created    → CoordinatorRequestDetails (with status "OPEN",
unitsCollected 0, unitsStillNeeded = unitsRequired)
```

Logs domain event via AppLogger.logRequestCreated(request.id, request.bloodType, request.unitsRequired, request.area.name, request.urgency).

Checks (in this exact order)

Part 1 — Format (CreateRequestDto):

1. patientReference is a non-empty string (trimmed)
2. bloodType is valid (@IsValidBloodType())
3. unitsRequired is an integer (@IsInt()) between 1 and 10 (@Min(1), @Max(10))
4. areaId is a non-empty string
5. urgency is one of 'CRITICAL', 'HIGH', 'NORMAL'
6. hospital is a non-empty string (trimmed)
7. note (optional) is a string

Part 2 — Rules (RequestsService.createRequest):

1. areaId exists in the Area table (400)
2. No existing OPEN request has the same patientReference (case-insensitive trimmed match) **and** the same bloodType (R-13, 409)

Errors

Problem	Status	Message
Missing patientReference	400	Patient reference is required.
Invalid bloodType	400	Invalid blood type. Must be one of O-, O+, A-, A+, B-, B+, AB-, AB+.
unitsRequired not an integer or outside 1–10	400	Units required must be a whole number between 1 and 10.
Missing areaId	400	Area is required.
Invalid urgency	400	Urgency must be CRITICAL, HIGH, or NORMAL.
Missing hospital	400	Hospital or blood-bank name is required.
Unknown areaId	400	Selected area is not supported.
Duplicate OPEN request with same patientReference and bloodType (R-13)	409	An open request for this patient reference and blood type already exists.

9. GET /requests & GET /requests/:id

Owner: **Adham Albasha**. Role: COORDINATOR.

9.1 GET /requests

Lists all blood requests for coordinators, supporting optional query filters (status, bloodType, areaId, urgency) (FR-7). Ordered by createdAt DESC.

```
GET
/api/requests?status=OPEN&bloodType=A-&areaId=area-homs&urgency=CRITIC
AL
Authorization: Bearer <coordinatorToken>

200 OK    → CoordinatorRequestSummary[] (section 4.4)
```

Problem	Status	Message
Invalid status query value	400	Invalid status filter. Must be OPEN, FULFILLED, or CANCELLED.
Invalid bloodType query value	400	Invalid blood type. Must be one of O-, O+, A-, A+, B-, B+, AB-, AB+.
Invalid urgency query value	400	Urgency must be CRITICAL, HIGH, or NORMAL.

9.2 GET /requests/:id

Returns full details of one blood request, including unitsRequired, unitsCollected, unitsStillNeeded, pledges, and donations (FR-13).

```
GET /api/requests/req-101
Authorization: Bearer <coordinatorToken>

200 OK    → CoordinatorRequestDetails (section 4.5)
```

Unknown request ID returns 404 Not Found with { "message": "Blood request not found." }.

10. PATCH /requests/:id & PATCH /requests/:id/cancel

Owner: **Adham Albasha**. Role: COORDINATOR.

10.1 PATCH /requests/:id

Updates an OPEN request's unitsRequired, urgency, and/or note (FR-6, R-9..R-14, R-18).

```
PATCH /api/requests/req-101
Authorization: Bearer <coordinatorToken>
Content-Type: application/json

{
  "unitsRequired": 2,
  "urgency": "HIGH",
  "note": "Updated unit count after cross-match."
}
```

```
200 OK    → updated CoordinatorRequestDetails
```

Rules checked in order (RequestsService.updateRequest)

1. The request exists (404 Blood request not found.).
2. The request status === 'OPEN' (R-9, R-14). If FULFILLED or CANCELLED, reject with 409.
3. If unitsRequired is provided, it must be >= request.unitsCollected (R-12). If unitsRequired < request.unitsCollected, reject with 400.
4. **Automatic Fulfillment on Reduction (R-11, R-18)**: If unitsRequired === request.unitsCollected (and unitsCollected > 0), inside prisma.$transaction, the request status automatically becomes 'FULFILLED', all remaining 'PLEDGED' pledges on this request are updated to 'RELEASED', and AppLogger.logRequestFulfilled(request.id) is called.

Problem	Status	Message
unitsRequired outside 1–10 or not an integer	400	Units required must be a whole number between 1 and 10.
Invalid urgency	400	Urgency must be CRITICAL, HIGH, or NORMAL.
Unknown request	404	Blood request not found.
Request is FULFILLED or CANCELLED (R-9, R-14)	409	Only OPEN requests can be updated.
unitsRequired < unitsCollected (R-12)	400	Units required cannot be reduced below the units already collected.

10.2 PATCH /requests/:id/cancel

Cancels an OPEN request with a required cancelReason (Flow E, FR-6, R-9, R-14, R-18).

```
PATCH /api/requests/req-101/cancel
Authorization: Bearer <coordinatorToken>
Content-Type: application/json

{
  "cancelReason": "Patient transferred to Damascus University
Hospital."
}

200 OK    → updated CoordinatorRequestDetails (status: "CANCELLED",
all PLEDGED pledges now "RELEASED")
```

Inside prisma.$transaction:

1. Updates the request status to 'CANCELLED' and stores cancelReason.
2. Updates all 'PLEDGED' pledges for this request to 'RELEASED' (R-18).
3. Logs the cancellation via AppLogger.logRequestCancelled(request.id, dto.cancelReason).

Problem	Status	Message
Missing or empty cancelReason	400	Cancellation reason is required.
Unknown request	404	Blood request not found.
Request is already FULFILLED or CANCELLED (R-9, R-14)	409	Only OPEN requests can be cancelled.

11. GET /requests/:id/eligible-donors

Owner: **Adham Albasha** (controller/route) & **Eyad Ibrahim** (MatchingService.getEligibleDonorsForRequest). Role: COORDINATOR.
Returns the ranked list of eligible donors for an OPEN request alongside the excluded-donor summary grouped by reason (Flow B, FR-9, R-1..R-8).

```
GET /api/requests/req-101/eligible-donors
Authorization: Bearer <coordinatorToken>

200 OK    → EligibleDonorsResponse (section 4.6)
```

How Matching, Exclusion Counting, and Ranking Work (R-1..R-8)

1. Verify the request exists (404) and its status === 'OPEN' (409 if FULFILLED or CANCELLED, R-9).
2. Query all donors whose areaId === request.areaId and whose bloodType is in BLOOD_COMPATIBILITY_MAP[request.bloodType] (R-1, R-6).
3. Evaluate each compatible same-area donor in this exact priority order to classify them as either **Eligible** or **Excluded** (R-6, R-8):
    - If !donor.user.isActive → increment excludedSummary.inactive
    - Else if isInRestPeriod(donor.lastDonationDate, ClockService.now()) → increment excludedSummary.restPeriod
    - Else if !donor.isAvailable → increment excludedSummary.unavailable
    - Else → donor is **Eligible**
4. Sort the **Eligible** donors according to **R-7**:
    - **Rule 1 (Exact match first)**: isExactBloodMatch = (donor.bloodType === request.bloodType) (true comes before false).
    - **Rule 2 (Longest since last donation / never donated first)**: Within each group, lastDonationDate === null comes first; otherwise older lastDonationDate (ASC) comes before newer lastDonationDate.
    - **Rule 3 (Alphabetical by name)**: Tie-breaker using donor.fullName.localeCompare(other.fullName).
5. Assign 1-based rank (1, 2, 3, ...) and return { requestId, bloodTypeNeeded, area, eligibleDonors, excludedSummary }.

Problem	Status	Message
Unknown request	404	Blood request not found.
Request is FULFILLED or CANCELLED (R-9)	409	Eligible donors can only be matched for OPEN requests.

12. POST /requests/:id/donations (Record a

Completed Donation)

Owner: **Adham Albasha** (with Eyad's MatchingService). Role: COORDINATOR.
Records a completed 1-unit donation for a donor on an OPEN request inside a **single database transaction** (Flow D, FR-12, R-19..R-22).

```
POST /api/requests/req-101/donations
Authorization: Bearer <coordinatorToken>
Content-Type: application/json

{
  "donorId": "dp-sara",
  "donationDate": "2026-11-01"
}
```

Success

```
201 Created    → updated CoordinatorRequestDetails (reflecting
incremented unitsCollected, completed pledge, new donation, and
FULFILLED status + released pledges if target reached)
```

Checks & Transaction Steps (in this exact order)

Part 1 — Format (RecordDonationDto):

1. donorId is a non-empty string
2. donationDate is a valid calendar date YYYY-MM-DD (@IsCalendarDate())

Part 2 — Pre-Transaction Verification (DonationsService.recordDonation):

1. Request exists (404 Blood request not found.).
2. Request status === 'OPEN' (409 Donations can only be recorded for OPEN requests. — R-9, R-19).
3. donationDate is not in the future compared to ClockService.now() (400 The donation date cannot be in the future. — R-21).
4. Donor exists (404 Donor not found.).
5. Donor is currently eligible for this request at the moment of recording via MatchingService.checkDonorEligibilityForRequest(donor, request, ClockService.now()) (R-19). If ineligible, call AppLogger.warnDonationRejected(requestId, donorId, check.reason) and throw 400 Bad Request with check.reason.

Part 3 — Atomic Database Transaction (prisma.$transaction, R-22):

All of the following operations execute inside a single prisma.$transaction(async (tx) => { ... }):

1. Create the Donation record (units: 1, donationDate, requestId, donorId, recordedById: coordinatorId) (R-20).
2. Increment request.unitsCollected by 1.
3. If the donor has an active 'PLEDGED' pledge on this request, update that pledge's status to 'COMPLETED'.
4. Update the donor's lastDonationDate to donationDate (or the newer of lastDonationDate and donationDate), starting their 56-day rest period (R-3, R-4).

5. If newUnitsCollected === request.unitsRequired:
    - Update request.status to 'FULFILLED' (R-11).
    - Update all remaining 'PLEDGED' pledges on this request to 'RELEASED' (R-18).
6. If any step inside the transaction throws an error, PostgreSQL rolls back the entire transaction so nothing is saved (R-22).

After the transaction commits, log AppLogger.logDonationRecorded(donation.id, requestId, donorId) (and AppLogger.logRequestFulfilled(requestId) if fulfilled).

Errors

Problem	Status	Message
Missing donorId	400	Donor ID is required.
Invalid donationDate format	400	Invalid date. Use the format YYYY-MM-DD.
Unknown request	404	Blood request not found.
Request is FULFILLED or CANCELLED (R-19)	409	Donations can only be recorded for OPEN requests.
donationDate is in the future (R-21)	400	The donation date cannot be in the future.
Unknown donor	404	Donor not found.
Donor account is inactive (R-6, R-19)	400	This donor account is deactivated and cannot donate.
Donor is marked unavailable (R-6, R-19)	400	This donor is currently marked as unavailable.
Donor is in rest period (R-6, R-19)	400	This donor is in rest period until YYYY-MM-DD.
Donor blood type is incompatible (R-1, R-6, R-19)	400	This donor's blood type (<donorType>) is not compatible with <neededType>.
Donor is in a different area (R-6, R-19)	400	This donor is in <donorArea>, but the request is in <requestArea>.

13. Donor Endpoints (/donors)

Owner: **Laith Haj Hosin** (with Eyad's MatchingService).

13.1 GET /donors/me

Role: DONOR. Returns the signed-in donor's profile, calculated eligibilityStatus, nextAllowedDonationDate, activePledge (if any), and donationHistory ordered by donationDate DESC (FR-3, FR-4, R-24).

```
GET /api/donors/me
Authorization: Bearer <donorToken>

200 OK    → DonorDashboardProfile (section 4.7)
```

13.2 PATCH /donors/me

Role: DONOR. Updates the signed-in donor's areaId, phone, and/or isAvailable flag (FR-3, R-24).

```
PATCH /api/donors/me
Authorization: Bearer <donorToken>
Content-Type: application/json

{
  "areaId": "area-homs",
  "phone": "0933112233",
  "isAvailable": false
}

200 OK    → updated DonorDashboardProfile
```

Problem	Status	Message
Invalid phone format	400	A valid phone number is required.
isAvailable is not a boolean	400	Availability must be a boolean value.
areaId does not exist in DB	400	Selected area is not supported.
phone already used by another user	409	An account with this phone number is already registered.

13.3 PATCH /donors/me/deactivate

Role: DONOR. Deactivates the signed-in donor's account (user.isActive = false, donorProfile.isAvailable = false) and automatically changes any active PLEDGED pledge held by the donor to WITHDRAWN inside a transaction (FR-3, R-24).

```
PATCH /api/donors/me/deactivate
Authorization: Bearer <donorToken>

200 OK    → { "message": "Your account has been deactivated." }
```

13.4 GET /donors/me/eligible-requests

Role: DONOR. Returns all OPEN requests that the signed-in donor is currently eligible to help with (MatchingService.checkDonorEligibilityForRequest returns eligible: true), ordered like the SOS board (CRITICAL → HIGH → NORMAL, then createdAt ASC) (Flow C, FR-10).

- If the donor is currently in their rest period, unavailable, or inactive, returns 200 OK with [].
- Each item includes canPledge: boolean, which is true when the donor does not already hold an active PLEDGED pledge and activePledgesCount < unitsStillNeeded.

```
GET /api/donors/me/eligible-requests
Authorization: Bearer <donorToken>
```

```
200 OK    → EligibleOpenRequest[] (section 4.8)
```

13.5 GET /donors

Role: COORDINATOR. Lists all donor profiles for coordinators (Section 5 Role Table: "Coordinator Can: view all donors"), ordered by fullName ASC.

```
GET /api/donors
Authorization: Bearer <coordinatorToken>

200 OK    → CoordinatorDonorItem[]
```

14. Pledge Endpoints (/requests/:id/pledges & /pledges/:id/withdraw)

Owner: **Laith Haj Hosin** (with Eyad's MatchingService). Role: DONOR.

14.1 POST /requests/:id/pledges

Creates an active PLEDGED pledge for the signed-in donor on an OPEN request (Flow C, FR-11, R-15..R-17).

```
POST /api/requests/req-101/pledges
Authorization: Bearer <donorToken>

201 Created    → DonorActivePledge (section 4.7)
```

Rules checked in this exact order (PledgesService.createPledge)

1. Request exists (404 Blood request not found.).
2. Request status === 'OPEN' (R-9, R-15). If FULFILLED or CANCELLED, throw 409.
3. Signed-in donor has no existing 'PLEDGED' pledge across **any** request (R-16). If they already hold an active pledge, throw 409.
4. Signed-in donor is eligible for this request via MatchingService.checkDonorEligibilityForRequest(donor, request, ClockService.now()) (R-6, R-15). If ineligible, throw 400 Bad Request with the exact reason (including nextAllowedDonationDate when in rest period: "You are in rest period until YYYY-MM-DD.").
5. The count of active 'PLEDGED' pledges on this request is strictly less than unitsStillNeeded (request.unitsRequired - request.unitsCollected) (R-17). If activePledgesCount >= unitsStillNeeded, throw 409.

Problem	Status	Message
Unknown request	404	Blood request not found.
Request is FULFILLED or	409	Pledges can only be made for

Problem	Status	Message
CANCELLED (R-9, R-15)		OPEN requests.
Donor already has an active PLEDGED pledge (R-16)	409	You already have an active pledge. You can hold at most one active pledge at a time.
Donor is in rest period (R-4, R-6, R-15)	400	You cannot pledge while in rest period. Your next allowed donation date is YYYY-MM-DD.
Donor is marked unavailable (R-6, R-15)	400	You cannot pledge while your availability is set to unavailable.
Donor blood type is incompatible (R-1, R-6, R-15)	400	Your blood type (<donorType>) is not compatible with this request (<neededType>).
Donor is in another area (R-6, R-15)	400	You can only pledge to requests in your registered area (<donorArea>).
Active pledges already equal unitsStillNeeded (R-17)	409	This request already has enough active pledges for all remaining units needed.

14.2 PATCH /pledges/:id/withdraw

Allows a donor to withdraw their own active PLEDGED pledge, setting its status to 'WITHDRAWN' (Flow C, FR-11, R-24).

```
PATCH /api/pledges/plg-501/withdraw
Authorization: Bearer <donorToken>

200 OK    → { "id": "plg-501", "status": "WITHDRAWN" }
```

Problem	Status	Message
Unknown pledge	404	Pledge not found.
Pledge belongs to another donor (R-24)	403	You can only withdraw your own pledge.
Pledge is not in PLEDGED status (COMPLETED, WITHDRAWN, or RELEASED)	409	Only active PLEDGED pledges can be withdrawn.

15. Calculated Values

The backend calculates these on every read. The frontend never calculates them:

```
unitsStillNeeded:
  Math.max(0, request.unitsRequired - request.unitsCollected)    (R-10)

nextAllowedDonationDate:
  lastDonationDate === null
    ? null                                                       (R-5)
    : addDays(lastDonationDate, REST_PERIOD_DAYS)                (R-3,
R-4, where REST_PERIOD_DAYS = 56)
```

```
isInRestPeriod:
  lastDonationDate !== null AND today < nextAllowedDonationDate (R-4)

eligibilityStatus (Donor):
  !user.isActive      → 'INACTIVE'
  isInRestPeriod      → 'REST_PERIOD'
  !donor.isAvailable  → 'UNAVAILABLE'
  otherwise           → 'ELIGIBLE'

isExactBloodMatch:
  donor.bloodType === request.bloodType                          (R-7)
```

16. Validation Rules

The **format** rules are checked by the DTO classes (class-validator). The **business** rules are checked by the services.

- **Unknown fields**: Rejected globally by ValidationPipe (whitelist: true, forbidNonWhitelisted: true) with 400 Bad Request — { "message": "Unknown field '<field>' is not allowed." }.
- **Phone**: Spaces and dashes are removed first via normalizePhone. Must match an optional + followed by 8 to 15 digits (0933112233 or +963933112233).
- **Blood Type**: Must be one of 'O-', 'O+', 'A-', 'A+', 'B-', 'B+', 'AB-', 'AB+'.
- **Units Required**: Must be an integer (1 to 10). Decimals (2.5), strings ("3"), 0, negative numbers, and numbers > 10 are rejected with 400.
- **Calendar Date (donationDate)**: YYYY-MM-DD and a real calendar date (2026-02-31 is rejected).

17. Errors, Logger & Swagger

Error Shape

Every error from every endpoint leaves the API in this exact shape (AllExceptionsFilter):
`{ "message": "Human-readable reason." }`

Status	Meaning
200	Success
201	Resource created (donor account, blood request, pledge, donation)
400	Invalid input format, unknown field, or eligibility/validation rule broken
401	Missing/invalid JWT or wrong sign-in credentials
403	Valid JWT, but role is not allowed or accessing another donor's resource

Status	Meaning
404	Request, donor, pledge, or endpoint not found
409	State conflict (duplicate phone/email, duplicate OPEN request, non-OPEN request, active pledge limit reached)
500	Unexpected server error ({ "message": "Something went wrong. Please try again." })

Custom Logger (AppLogger)

Per Section 12 ("Logger: log meaningful events — request created, cancelled and fulfilled, donation recorded, and rejected donation attempts with the reason. Do not log every request, and never log passwords or tokens"):

- REQUEST_CREATED: logged when POST /api/requests succeeds
- REQUEST_CANCELLED: logged when PATCH /api/requests/:id/cancel succeeds
- REQUEST_FULFILLED: logged when unitsCollected reaches unitsRequired (via donation or request update)
- DONATION_RECORDED: logged when POST /api/requests/:id/donations commits
- DONATION_REJECTED: logged at WARN level when POST /api/requests/:id/donations is rejected due to an ineligible donor, non-open request, or future date, including the exact rejection reason

Swagger (/api/docs)

Every controller uses @ApiTags(...), every endpoint uses @ApiOperation({ summary: '...' }) and @ApiResponse(...), every protected endpoint uses @ApiBearerAuth(), and every DTO property uses @ApiProperty(...) / @ApiPropertyOptional(...).

18. Frontend API Service

```
client/src/services/
├── http.ts
├── auth.api.ts
├── donors.api.ts
├── pledges.api.ts
└── requests.api.ts
```

Functions:

```
// auth.api.ts (Molham)
getAreas(): Promise<Area[]>
signUpDonor(payload: SignUpPayload): Promise<AuthResponse>
signIn(payload: SignInPayload): Promise<AuthResponse>
getMe(): Promise<AuthUser>

// donors.api.ts (Taymaa)
getMyDonorProfile(): Promise<DonorDashboardProfile>
updateMyDonorProfile(payload: UpdateDonorProfilePayload):
Promise<DonorDashboardProfile>
```

```
deactivateMyAccount(): Promise<{ message: string }>
getMyEligibleRequests(): Promise<EligibleOpenRequest[]>
getAllDonors(): Promise<CoordinatorDonorItem[]>

// pledges.api.ts (Taymaa)
createPledge(requestId: string): Promise<DonorActivePledge>
withdrawPledge(pledgeId: string): Promise<{ id: string; status:
PledgeStatus }>

// requests.api.ts (Rami)
getSosRequests(): Promise<PublicSosRequest[]>
getRequests(filters?: RequestFilters):
Promise<CoordinatorRequestSummary[]>
getRequestById(requestId: string): Promise<CoordinatorRequestDetails>
createRequest(payload: CreateRequestPayload):
Promise<CoordinatorRequestDetails>
updateRequest(requestId: string, payload: UpdateRequestPayload):
Promise<CoordinatorRequestDetails>
cancelRequest(requestId: string, payload: CancelRequestPayload):
Promise<CoordinatorRequestDetails>
getEligibleDonors(requestId: string): Promise<EligibleDonorsResponse>
recordDonation(requestId: string, payload: RecordDonationPayload):
Promise<CoordinatorRequestDetails>
```

Only the three React Contexts (AuthContext, DonorContext, RequestsContext) call these service functions. Pages and components use the context hooks.

19. Testing Checklist (api/requests/api.http)

Owner: **Laith Haj Hosin**. api/requests/api.http includes one runnable request for every item below (covering Sections 13 and 19 of the specification):

- **Areas & Public SOS Board**:
    1. GET /api/areas returns seeded areas (200).
    2. GET /api/requests/sos returns OPEN requests ordered by CRITICAL → HIGH → NORMAL then oldest first, with zero personal data or patient references (200).
- **Auth & Roles**: 3. POST /api/auth/sign-up valid donor registration (201) and immediate GET /api/donors/me (200). 4. POST /api/auth/sign-up with missing field, unknown blood type, or unknown area (400). 5. POST /api/auth/sign-up with duplicate phone or email (409). 6. POST /api/auth/sign-in as Coordinator (200) and as Donor (200). 7. POST /api/auth/sign-in with wrong password (401 with generic message). 8. Protected endpoint called without a token (401). 9. Donor calling a coordinator-only operation (POST /api/requests or GET /api/donors) (403).
- **Requests & Lifecycle**: 10. POST /api/requests valid A- request in Homs for 3 units (201). 11. POST /api/requests with unitsRequired outside 1–10 or non-integer (400), or with unknown field (400). 12. POST /api/requests second OPEN request with same patientReference and bloodType (409). 13. GET /api/requests with filters (status, bloodType, areaId, urgency) (200). 14. PATCH /api/requests/:id reducing unitsRequired

below unitsCollected (400). 15. PATCH /api/requests/:id and PATCH /api/requests/:id/cancel on a FULFILLED or CANCELLED request (409).

- **Matching & Eligible Donors**: 16. GET /api/requests/:id/eligible-donors for an A- request in Homs returns only active, available, non-resting A- and O- donors in Homs, ordered with A- before O-, oldest donation / never donated first, then by name, plus excludedSummary (restPeriod, unavailable, inactive) (200).
- **Donor Profile & Pledges**: 17. POST /api/requests/:id/pledges by an eligible donor (201). 18. POST /api/requests/:id/pledges by a donor who already holds an active pledge (409). 19. POST /api/requests/:id/pledges by a donor in rest period (400 including exact nextAllowedDonationDate), unavailable (400), incompatible blood type (400), or in another area (400). 20. POST /api/requests/:id/pledges when active pledges already equal unitsStillNeeded (409). 21. PATCH /api/pledges/:id/withdraw withdraws the pledge (200).
- **Donations & Transactions**: 22. POST /api/requests/:id/donations for an eligible donor increases unitsCollected by 1, marks the donor's pledge COMPLETED, and sets nextAllowedDonationDate to donationDate + 56 days (201). 23. POST /api/requests/:id/donations for the final required unit automatically marks the request FULFILLED and sets all remaining PLEDGED pledges to RELEASED (201). 24. POST /api/requests/:id/donations for an ineligible donor (e.g., now in rest period) or with a future donationDate is rejected (400), logs a warning, and leaves the database unchanged. 25. Request or donor ID that does not exist returns 404.

20. Changing the Contract

When an endpoint, DTO field, or response shape needs to change:

1. Ask the team lead (Eyad Ibrahim) first.
2. Update the backend implementation and Swagger decorators.
3. Update this docs/API.md.
4. Update client/src/services/ and client/src/types/.
5. Update docs/PROJECT-STRUCTURE.md if any type or method signature changed.
6. Notify the affected frontend/backend pair in the team chat.

Never change a response shape, a field name, or an error message silently.